

Customer Behavior Analysis Report

Project Overview

This report presents an analysis of customer behavior using transactional data of approximately 3,900 customers. The objective is to understand purchasing patterns, customer segmentation, and key drivers of revenue using SQL and Power BI.

The dataset includes demographic, transactional, and behavioral attributes such as age, gender, purchase amount, subscription status, discount usage, and purchase frequency.

Data Preparation

Rows: 3,900 - Columns: 18 - Key Features: - Customer demographics (Age, Gender, Location, Subscription Status) - Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color) - Shopping behavior (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type) - Missing Data: 37 values in Review Rating column

- Cleaned dataset and ensured no missing values
- Standardized categorical fields
- Created derived column: Age Group
- Verified data types

Exploratory Data Analysis using Python

- We began with data preparation and cleaning in Python:
- Data Loading: Imported the dataset using pandas.
- Initial Exploration: Used `df.info()` to check structure and `describe()` for summary statistics.

[9]: df.describe(include = 'all')

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	3900	3900
unique	NaN	NaN	2	25	5	NaN	50	4	25	4	NaN	2	6	2	2
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	No	No
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	2223	2223
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	NaN	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	NaN	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN	NaN

- Missing Data Handling:
Checked for null values and imputed missing values in the Review Rating column using the median rating of each product category.
- Column Standardization:
Renamed columns to snake_case for better readability and consistency.

Before	After
Customer ID	customer_id
Age	age
Gender	gender
Item Purchased	item_purchased
Category	category
Purchase Amount (USD)	purchase_amount
Location	location
Size	size
Color	color
Season	season
Review Rating	review_rating
Subscription Status	subscription_status
Shipping Type	shipping_type
Discount Applied	discount_applied
Promo Code Used	promo_code_used
Previous Purchases	previous_purchases
Payment Method	payment_method

- **Feature Engineering:**
Created age_group column by binning customer ages into categories.
Created **purchase_frequency_days** column from purchase data.
- **Data Consistency Check:**
Verified if discount_applied and promo_code_used were redundant; dropped **promo_code_used** column.
- **Database Integration:**
Connected Python script to **PostgreSQL** and loaded the cleaned DataFrame into the database for SQL analysis.

Key KPIs

\$59.76 Average Purchase Amount	3.75 Average Review Rating	3.9K Number of customers	233K Total Revenue
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Total Revenue: 233K

Total Customers: 3.9K

Average Purchase Amount: \$59.76

Average Rating: 3.75

Key Business Insights

4.1 Customer Demographics

- Women contribute a larger share of purchases
- Young Adults generate the highest revenue

4.2 Category Performance

- Clothing category generates highest revenue
- Accessories follow as second-highest contributor

4.3 Customer Segmentation

- Loyal customers: 80%
- Returning customers: 18%
- New customers: 2%

4.4 Subscription Behavior

- 73% customers are non-subscribers
- Subscribers contribute significantly to revenue

4.5 Discount Impact

- Customers using discounts still maintain similar spending
- Discounts influence purchasing decisions but not drastically increase spend

4.6 Shipping Behavior

- Express and fast shipping users tend to spend slightly more
- Standard shipping remains most commonly used

4.7 Purchase Patterns

- Frequent buyers contribute higher revenue
- Repeat customers show higher loyalty and engagement

SQL Analysis Summary & Business Transactions

Performed analysis on revenue, discounts, segmentation, category performance, and payment behavior using SQL queries.

We performed structured analysis in PostgreSQL to answer key business questions:

1. Revenue by Gender – Compared total revenue generated by male vs. female customers.

	gender text	revenue numeric
1	Female	75191
2	Male	157890

2. Top 5 Products by Rating – Found products with the highest average review ratings.

	item_purchased text	Average Product Rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

3. Shipping Type Comparison – Compared average purchase amounts between Standard and Express shipping.

	shipping_type text	round numeric
1	Standard	58.46
2	Express	60.48

4. Subscribers vs. Non-Subscribers – Compared average spend and total revenue across subscription status.

	subscription_status text	total_customers bigint	avg_spend numeric	total_revenue numeric
1	Yes	1053	59.49	62645.00
2	No	2847	59.87	170436.00

5. Discount-Dependent Products – Identified 5 products with the highest percentage of discounted purchases.

	item_purchased text	discount_rate numeric
1	Hat	50.00
2	Sneakers	49.66
3	Coat	49.07
4	Sweater	48.17
5	Pants	47.37

6. Customer Segmentation – Classified customers into New, Returning, and Loyal segments based on purchase history.

	segment text	count bigint
1	Loyal	3116
2	New	83
3	Returning	701

7. Top 3 Products per Category – Listed the most purchased products within each category.

	item_rank bigint	category text	item_purchased text	total_orders bigint
1	1	Accessori...	Jewelry	171
2	2	Accessori...	Sunglasses	161
3	3	Accessori...	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	149
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161
12	1	python -m	Shoes	1

8. Revenue by Age Group – Calculated total revenue contribution of each age group.

	age_group text	total_revenue numeric
1	Young Adult	62143
2	Middle-aged	59197
3	Adult	55978
4	Senior	55763

9. Age group spends the most per order

	age_group text	avg_spend numeric
1	Young Adult	60.45
2	Middle-aged	60.04
3	Adult	59.42
4	Senior	59.07

10. Category generates highest revenue

	category text	category_revenue numeric
1	Clothing	104264
2	Accessori...	74200
3	Footwear	36059
4	Outerwear	18524
5	python -m	34

11. Top 10 location generates the highest revenue

	location text	total_revenue numeric
1	Montana	5784
2	Illinois	5617
3	California	5605
4	Idaho	5587
5	Nevada	5514
6	Alabama	5261
7	New York	5257
8	North Dako...	5220
9	West Virgi...	5174
10	Nebraska	5172

12. Size is most popular in each category

	category text	size text	total_orders bigint
1	Accessori...	M	562
2	Clothing	M	778
3	Footwear	M	267
4	Outerwear	M	148
5	python -m	L	1

13. Top 10 colors are most purchased

	color text	total_orders bigint	total_revenue numeric
1	Olive	177	10292
2	Yellow	174	10308
3	Silver	173	9832
4	Teal	172	10459
5	Green	169	11104
6	Black	167	9753
7	Cyan	166	10274
8	Violet	166	10245
9	Gray	159	9936
10	Maro...	158	9405

14. Items have high rating but low revenue

	item_purchased text	avg_rating numeric	total_revenue numeric	total_orders bigint
1	Jeans	3.65	7548	124
2	Gloves	3.86	8477	140
3	Sneakers	3.76	8635	145
4	Backpack	3.75	8636	143
5	Hoodie	3.72	8767	151
6	Handbag	3.78	8857	153
7	Boots	3.82	9018	144
8	Sandals	3.84	9200	160
9	Shoes	3.75	9240	150
10	T-shirt	3.78	9248	147
11	Jacket	3.76	9249	163
12	Socks	3.75	9252	159
13	Coat	3.73	9275	161
14	Hat	3.80	9375	154
15	Skirt	3.78	9402	158
16	Shorts	3.71	9433	157
17	Sweater	3.76	9462	164
18	Scarf	3.71	9561	157
19	Belt	3.76	9635	161
20	Sunglasses	3.75	9649	161

Business Recommendations

- Focus on customer acquisition (new users are low)
- Promote subscription plans to increase retention
- Optimize top-performing categories (Clothing)
- Use targeted discounts instead of blanket offers
- Improve customer experience for high-value segmen

7. Conclusion

This project demonstrates how customer data can be transformed into actionable insights. By combining SQL analysis with Power BI visualization, key patterns in customer behavior, spending habits, and business performance were identified.

